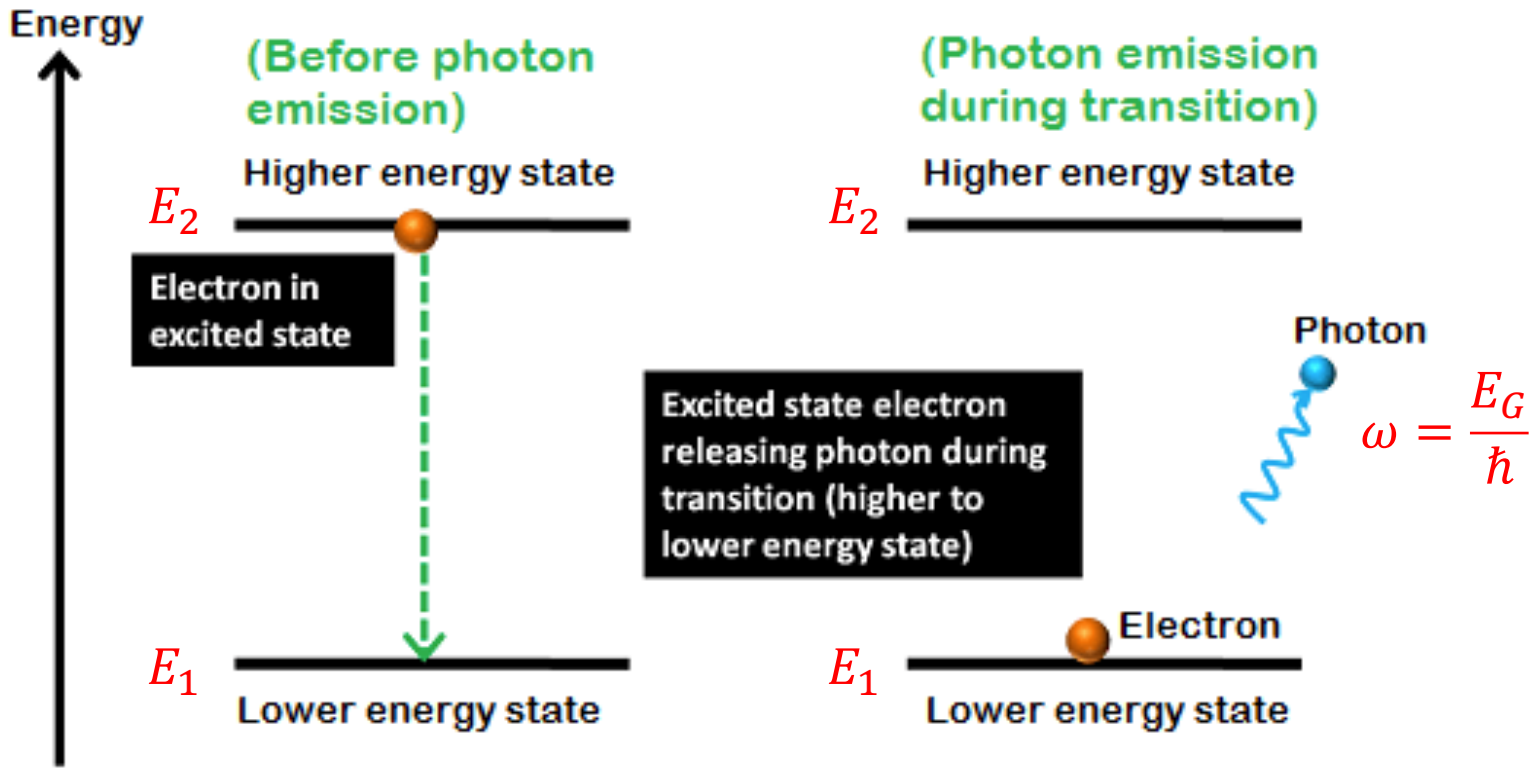
Light Emitting Diodes

The following subjects will be covered

- The PN Junction Band Diagram.
- LED Basic Design and Operation.
- LED Efficiency.
- Improved LED Designs.
- Basic LED Types.
- LED Characterization.
- LED Materials.

Photon Generation and Absorption

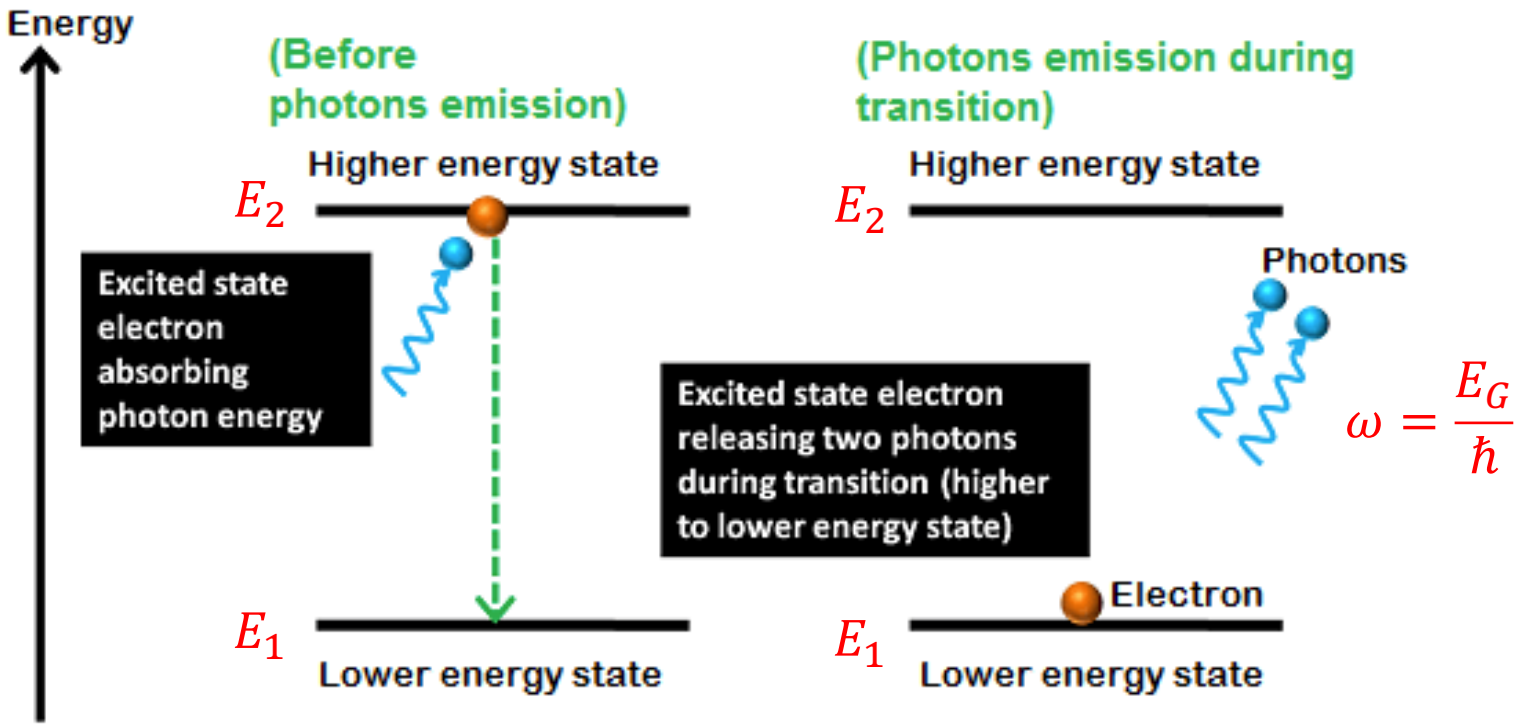
Spontaneous Emission



Electron-hole Re-combination (radiative/nonradiative)

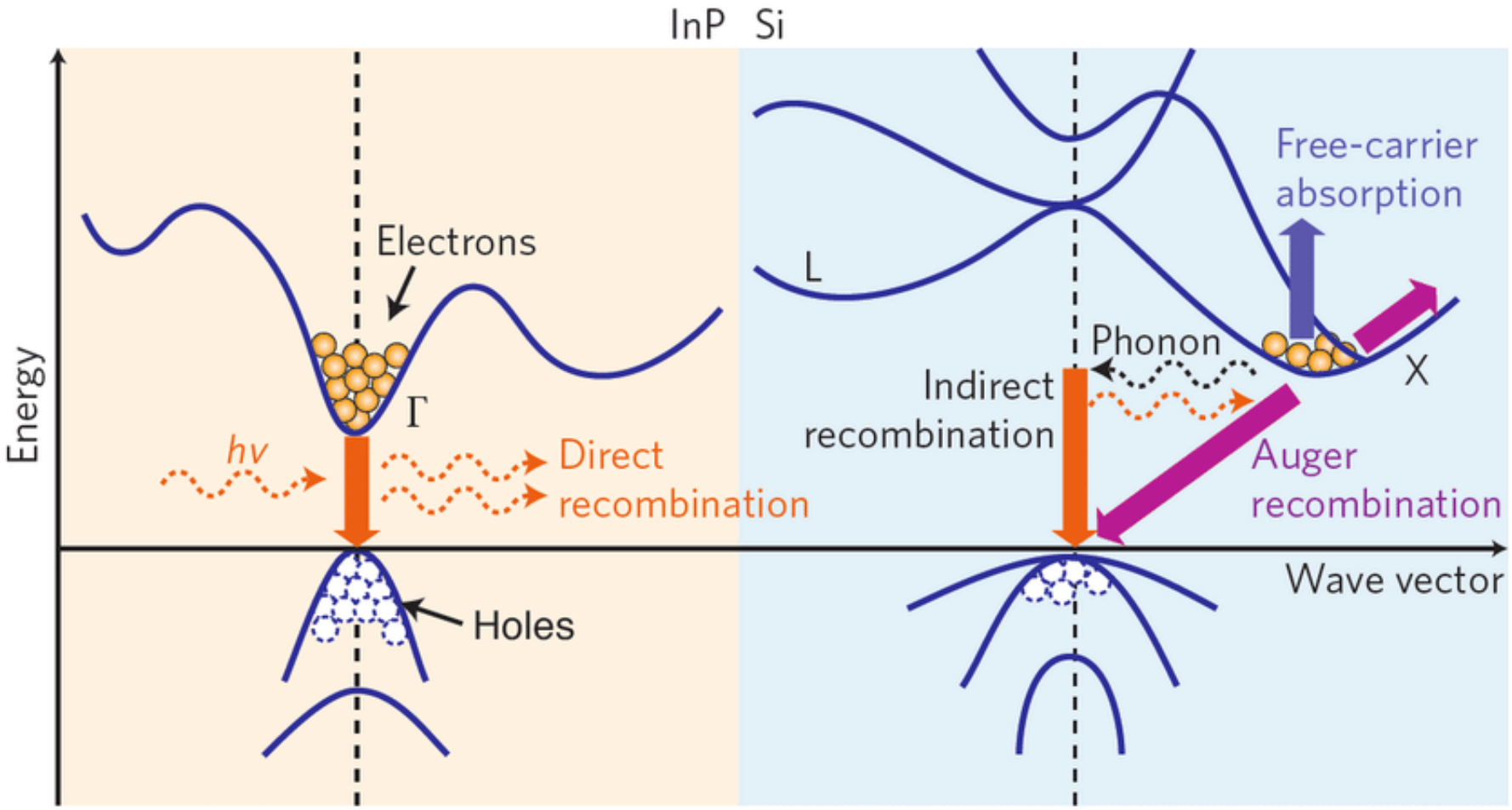
Photon Generation and Absorption

Stimulated Emission



Electron-hole
Re-combination
(radiative/nonradiative)

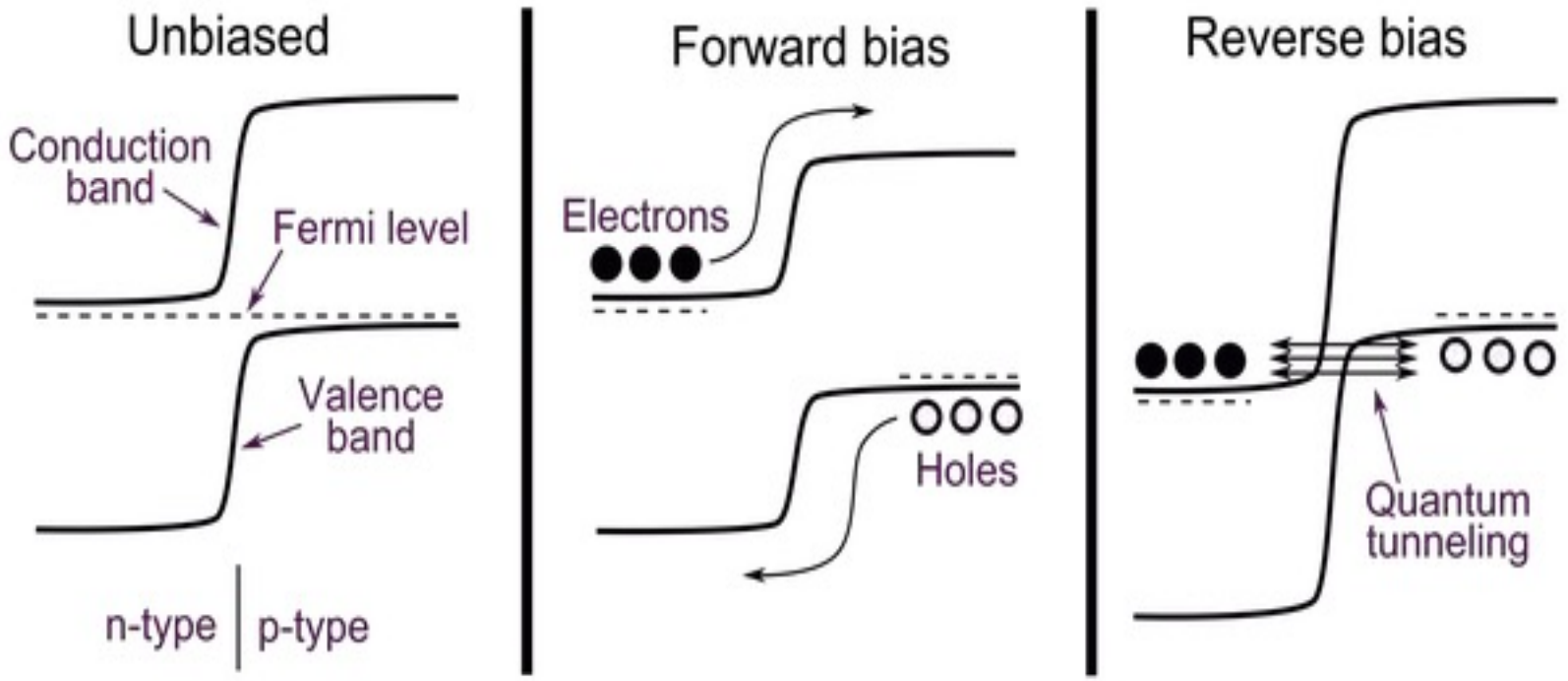
Direct and Indirect Gap Materials



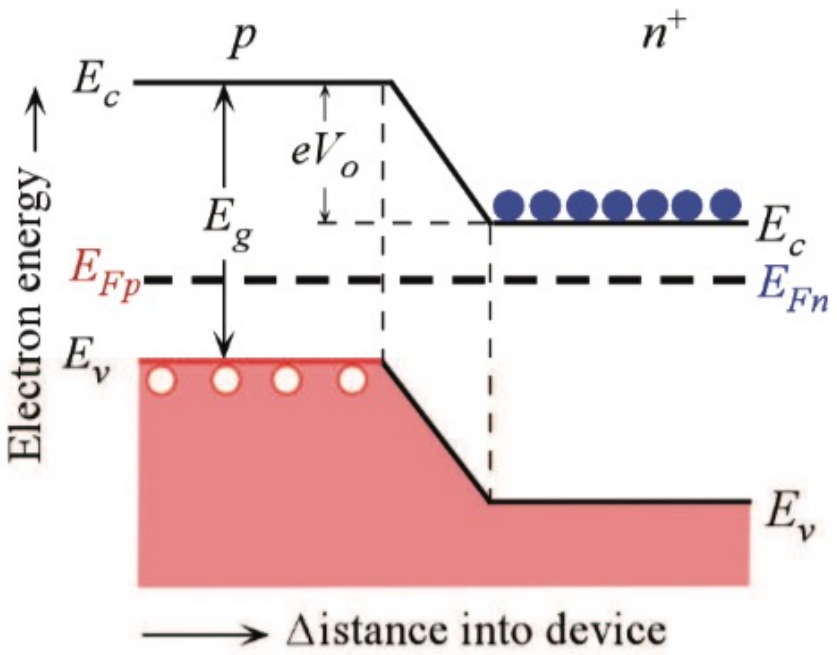
Direct and Indirect Gap Materials

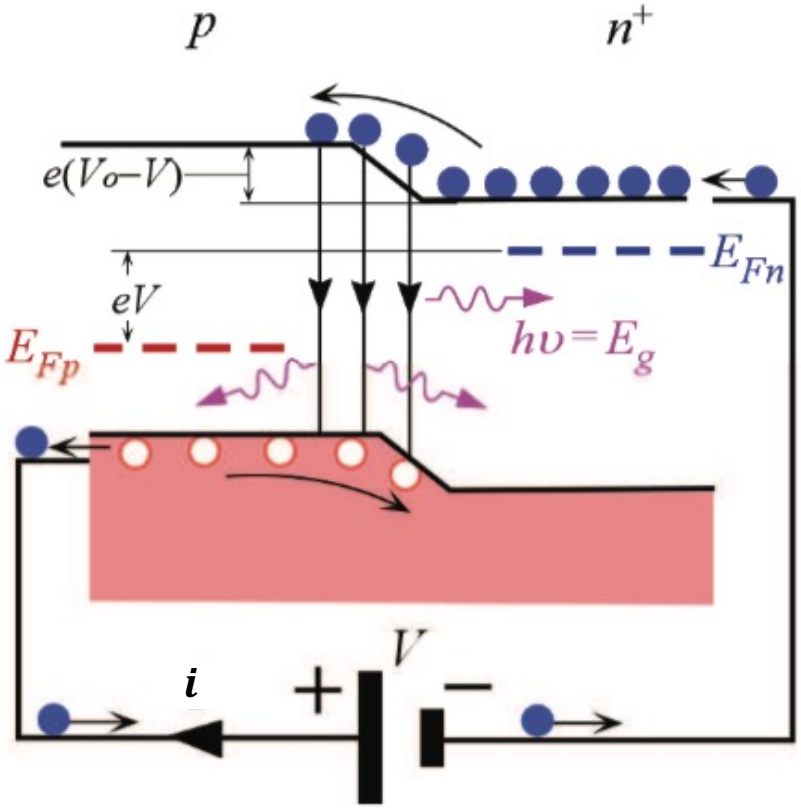
	Material	Direct / IndirectBandgap	Band Gap Energy at 300 K (eV)
Elements	C (di	Indirect	5.47
	Ge	Indirect	0.66
	Si	Indirect	1.12
	Sn	Direct	0.08
Groups III-V compounds	GaAs	Direct	1.42
	InAs	Direct	0.36
	InSb	Direct	0.17
	GaP	Indirect	2.26
	GaN	Direct	3.36
	InN	Direct	0.70
	InP	Direct	1.344
Groups IV-IV compounds	α-SiC	Indirect	2.99
Groups II-VI compounds	ZnO	Direct	3.35
	CdSe	Direct	1.70
	ZnS	Direct	3.68

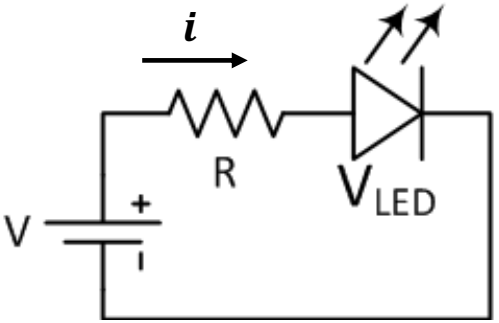
The Biased PN Junction Band Diagram



LED Basic Operation

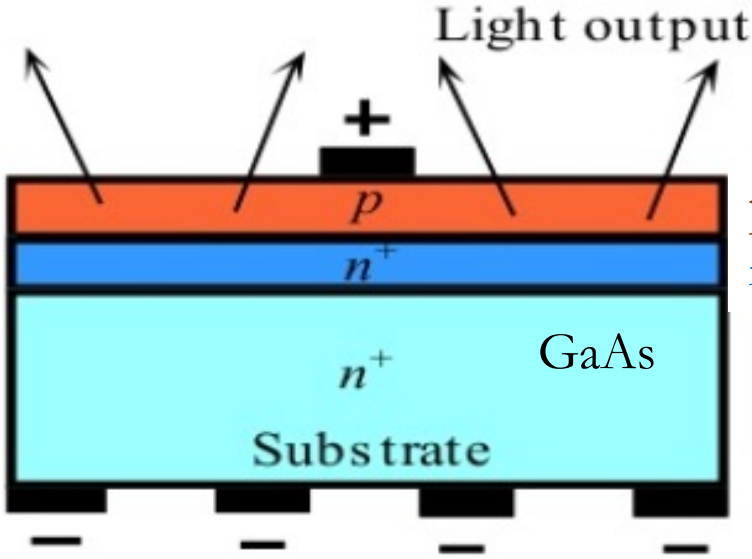






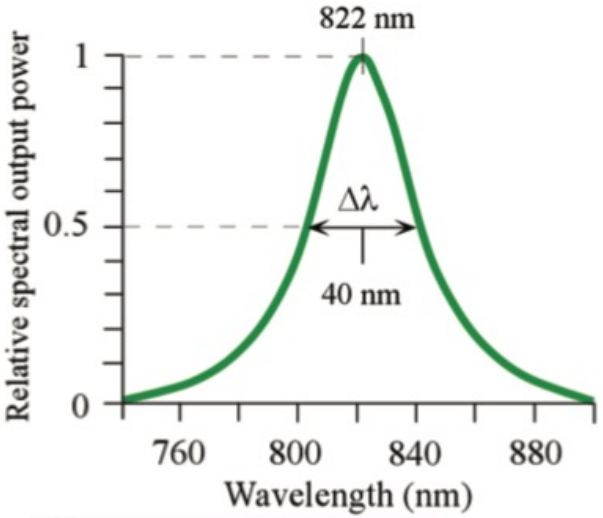
Input electrical power = Vi

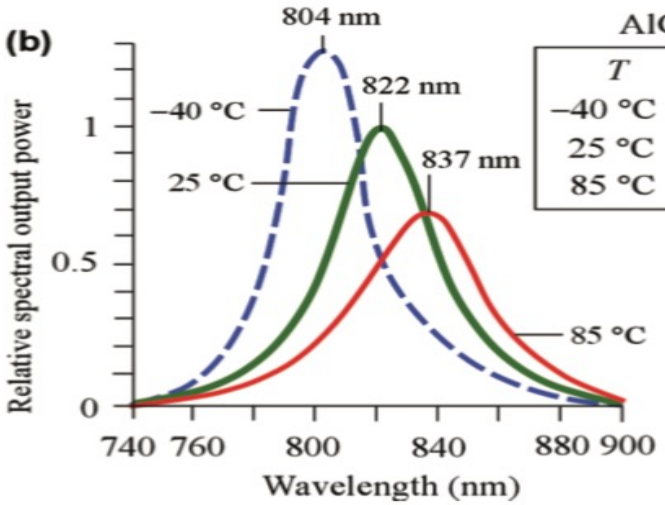
LED Basic Structure



p-AlGaAs

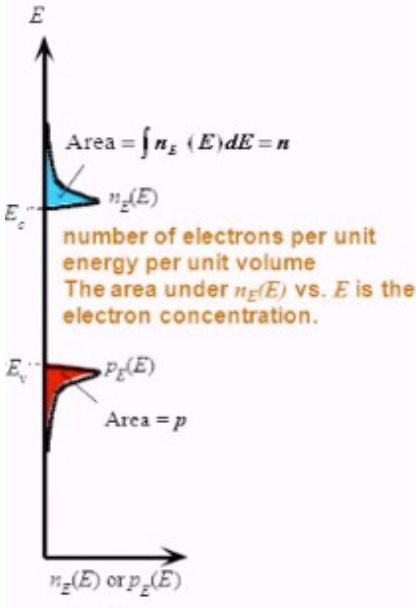
n-AlGaAs



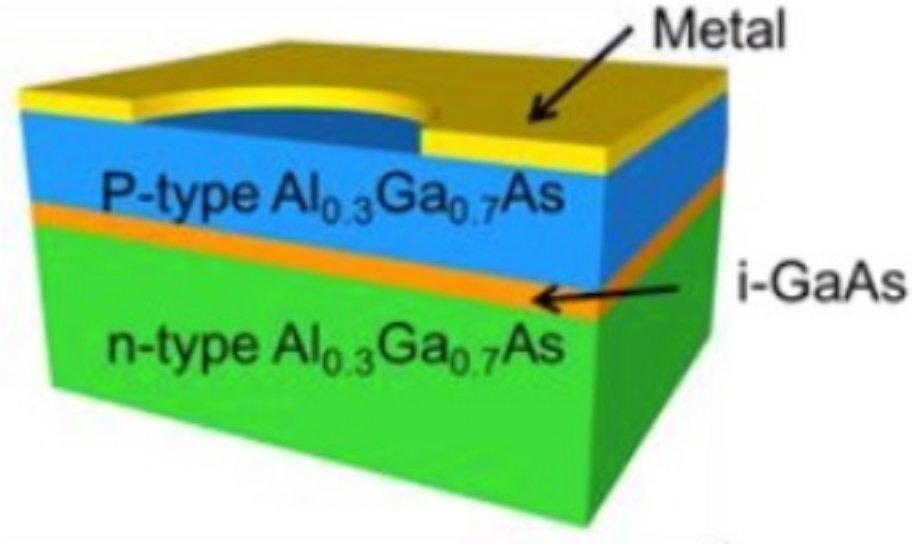


AlGaAs IR LED

T	λ_o	$\Delta\lambda$
−40 °C	804 nm	30 nm
25 °C	822 nm	40 nm
85 °C	837 nm	48 nm



LED Basic Structure

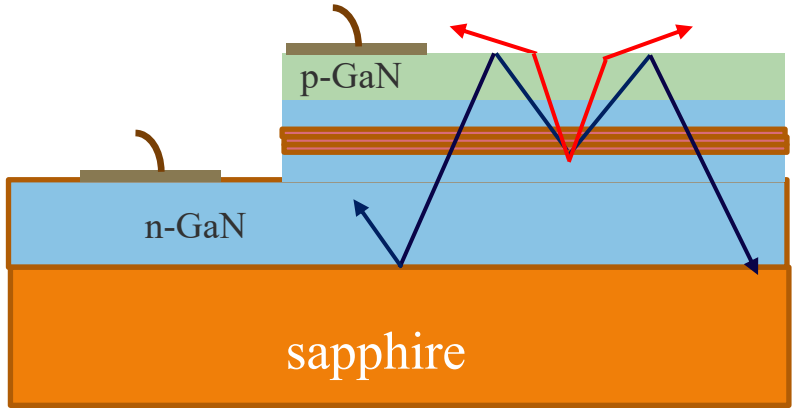


	Unit	GaAs	$Al_xGa_{1-x}As$, $0<x<0.45$
Bandgap Energy	eV	1.424	1.424 + 1.247x
Electron Effective Mass	m_0	0.067	0.067 + 0.083x
Hole Effective Mass	m_0	0.5	0.5 + 0.29x
Dielectric Constant	ε_0	13.1	13.1 – 3x
Conduction Band Discontinuity	%	-	$\Delta E_C \sim 67\% \Delta E_g$
Valence Band Discontinuity	%	-	$\Delta E_V \sim 33\% \Delta E_g$

LED Efficiency

The basic pn design of an LED suffers from several problems that limit its efficiency.

$$\eta = \frac{\text{Output Optical Power}}{\text{Input Electrical power}} = \frac{P_o}{I\,V}$$



- Carrier confinement: carries escape the depletion region which results in a loss of radiative recombination (injection loss).
- Optical confinement: generated photons are scattered in all directions.
- Non-radiative recombination: some carries recombine in a non-radiative fashion, because of traps, defects, etc.
- Light escape cone and total internal reflection: light is trapped inside the device.
- Absorption loss: light attenuation inside material.

LED Efficiency

Convert all EHP to Photons

- Non-radiative recombination
- Induced polarization
- Current Crowding
- Ohmic loss
- Electron leakage
- Thermal effects

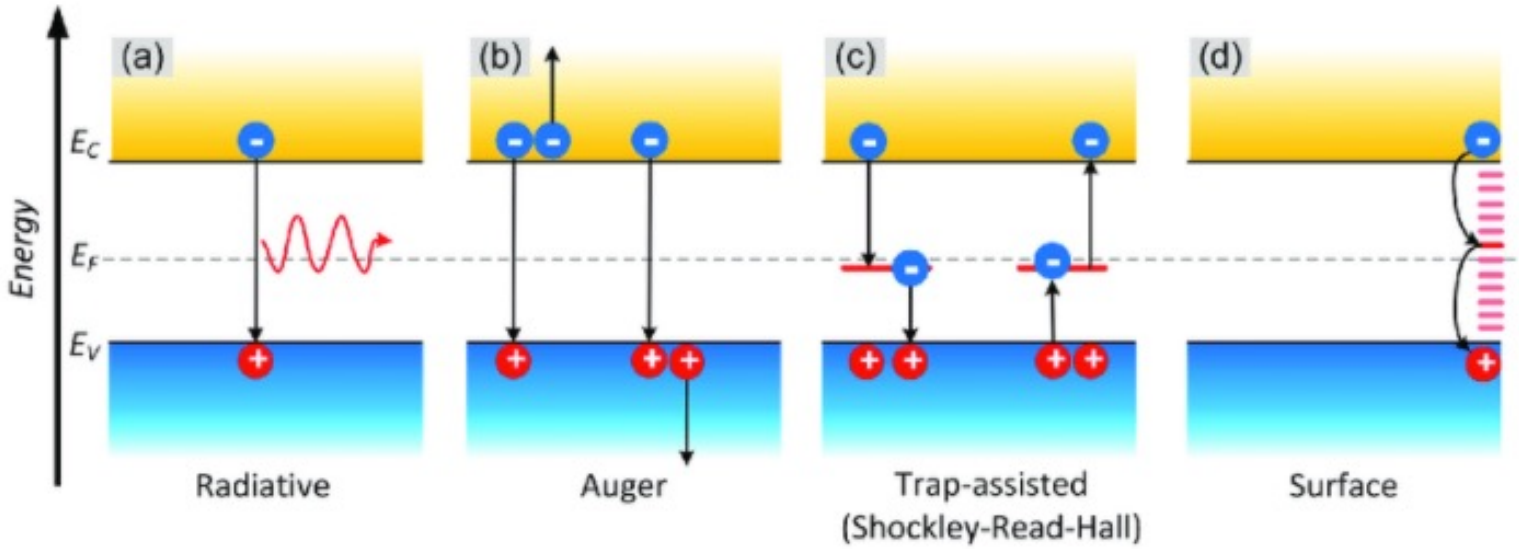
Save all Photons

- Fresnel loss
- Total internal reflection loss
- Absorption loss.
- Packaging
- Radiation pattern

Interdependency

- Extraction enhancement is very crucial for devices with low IQE.
- Positive and negative interdependence.

LED Efficiency: Nonradiative Recombination



Auger Recombination

Shockley-Read-Hall (SRH)

Surface states Defects

Internal quantum efficiency $\eta_i = \dfrac{R_r}{R_r + R_{nr}}$

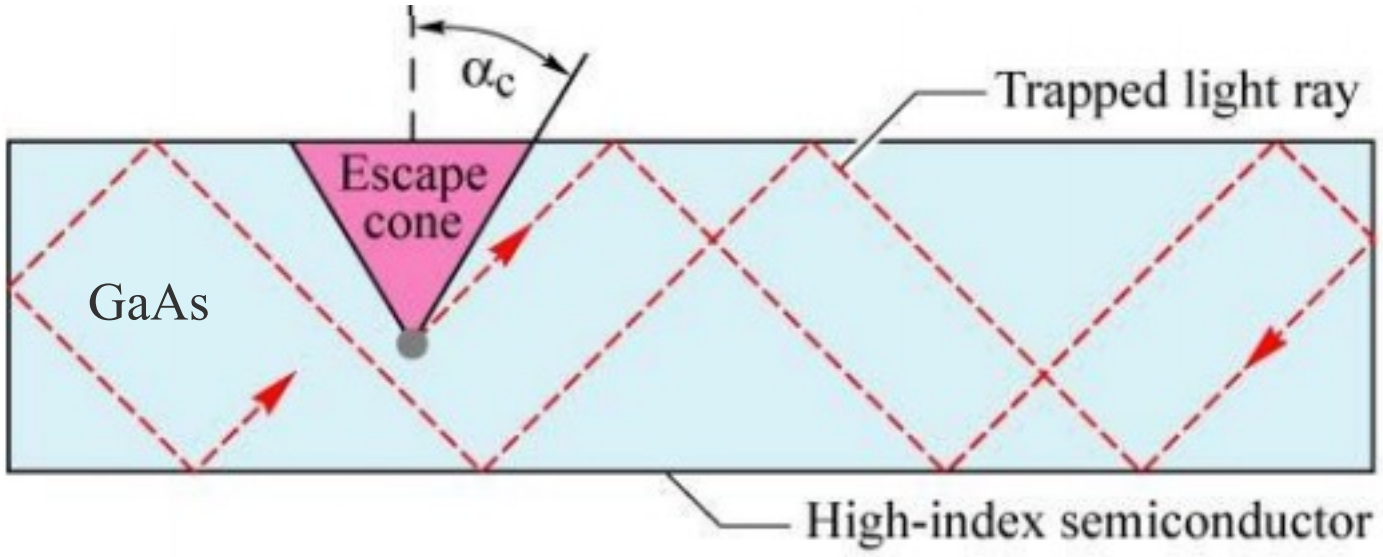
R: Recombination rate

$$\frac{1}{\tau} = \frac{1}{\tau_r} + \frac{1}{\tau_{nr}}$$

$$\eta_i = \frac{\tau}{\tau_r}$$

τ : Recombination lifetime

LED Efficiency: Light Escape Cone and TIR



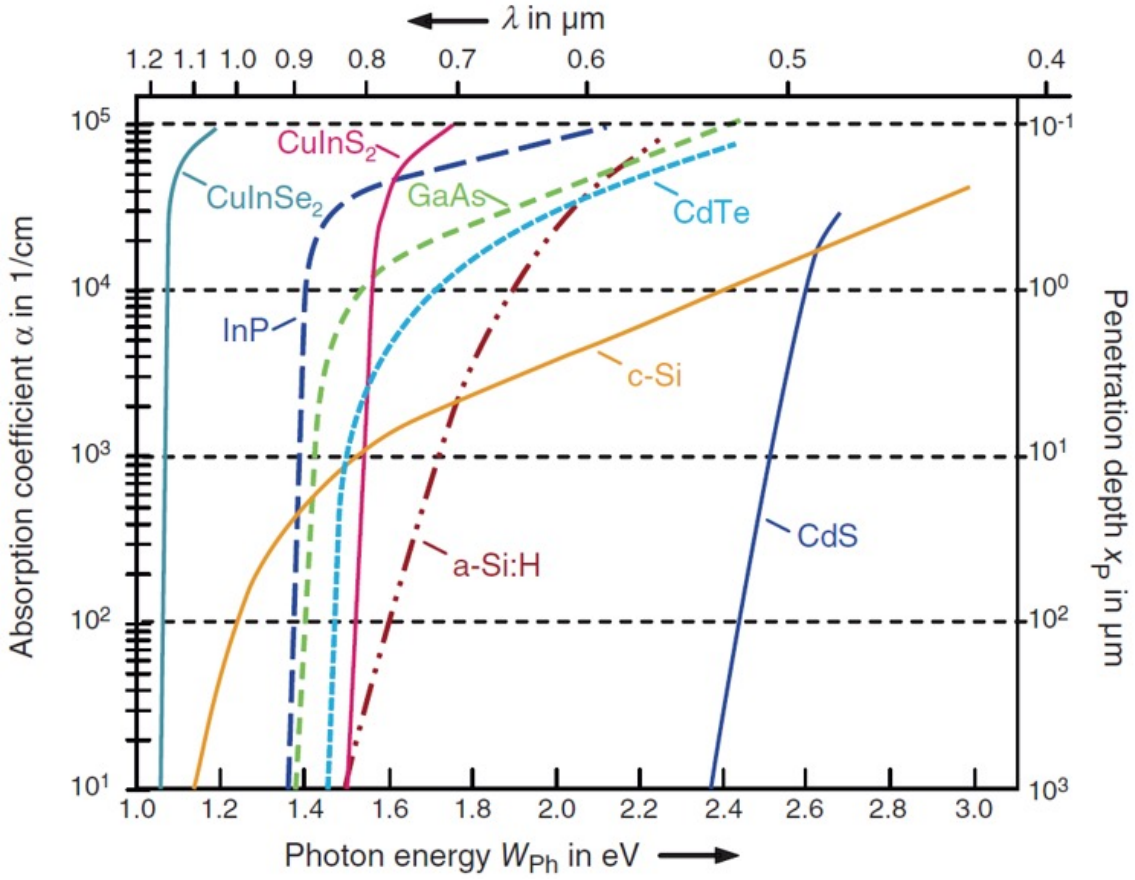
Fresnel loss: $R > \left(\frac{n_2 - n_1}{n_2 + n_1}\right)^2 = 0.32$

η_1

TIR loss: $\alpha_c = \sin^{-1} n_2/n_1 = 16^o$

η_2

LED Efficiency: Absorption



Attenuation loss: $e^{-\alpha\, l}$

η_3

LED Efficiency

Internal quantum efficiency $\eta_i = \dfrac{Photon\ flux}{electron\ flux} = \dfrac{\varphi}{i/q}$

Extraction efficiency $\eta_e = \eta_1 \times \eta_2 \times \eta_3$

External quantum efficiency $\eta_{ex} = \eta_i \times \eta_e$

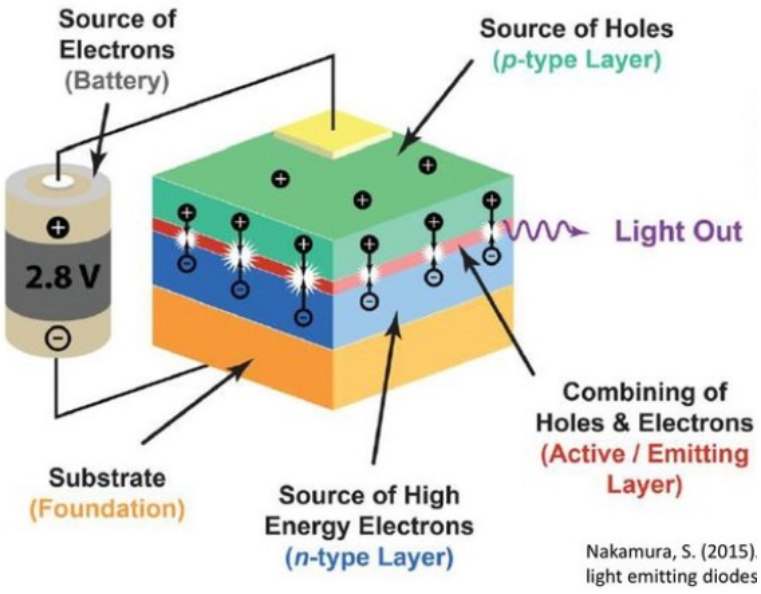
Photon flux $\varphi = \eta_i \dfrac{i}{q}$

Output photon flux $\varphi_o = \eta_e\, \varphi = (\eta_1 \times \eta_2 \times \eta_3 \times \eta_i) \dfrac{i}{q}$

$= \eta_{ex} \dfrac{i}{q}$

Output optical power $P_o = \hbar\omega\, \varphi_o = \hbar\omega\, \eta_{ex} \dfrac{i}{q}$

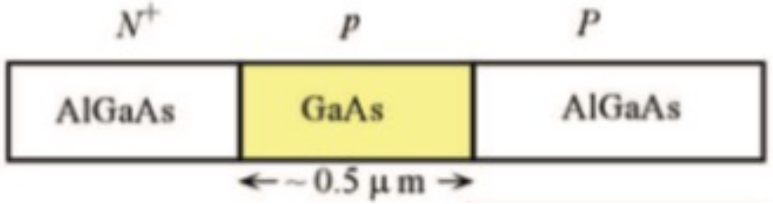
Wall-plug efficiency $\eta = \dfrac{P_o}{i\, V} = \eta_{ex} \dfrac{\hbar\omega}{q\, V}$

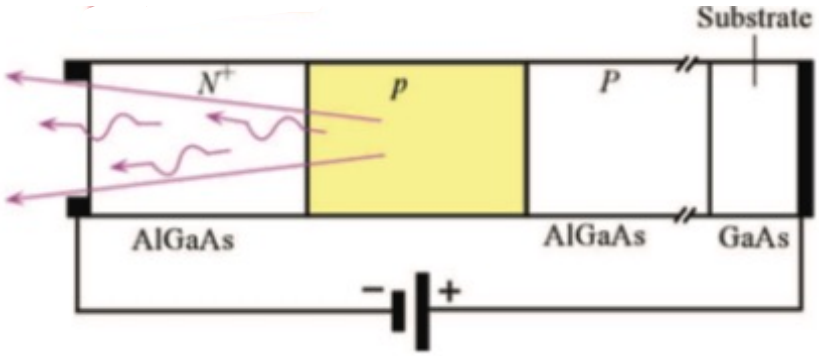


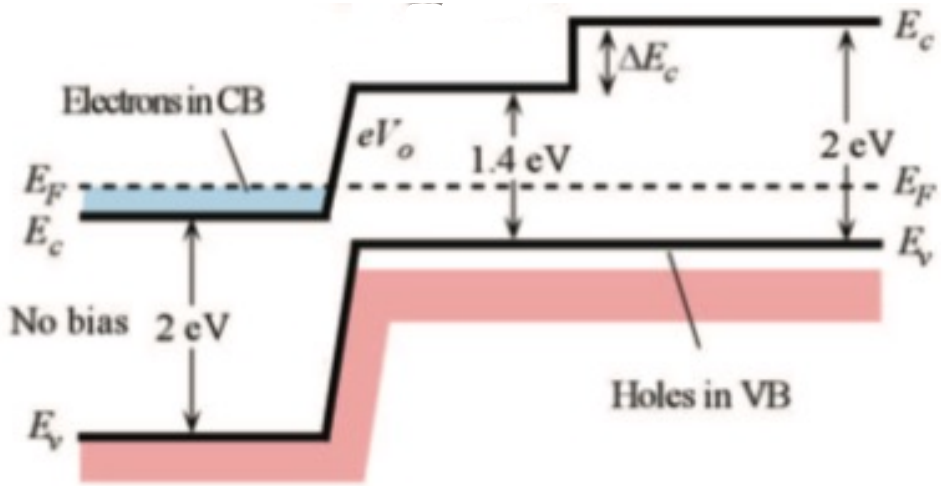
Basic LED design has a wall-plug efficiency of around 2-20%

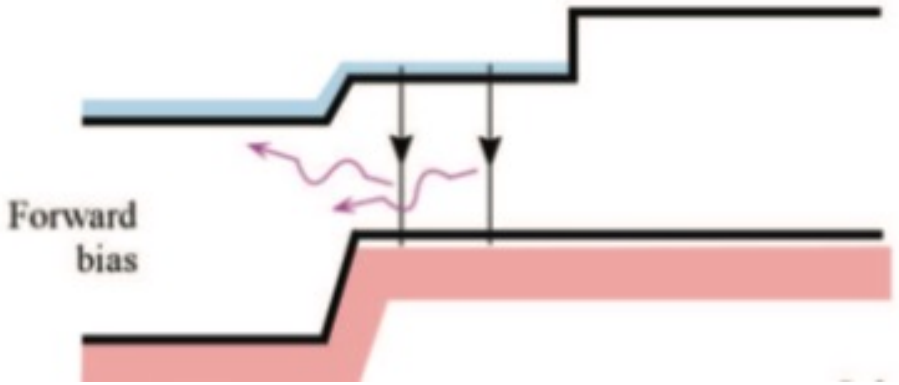
Improved LED Designs: Heterojunctions

Improves carrier and optical confinement.



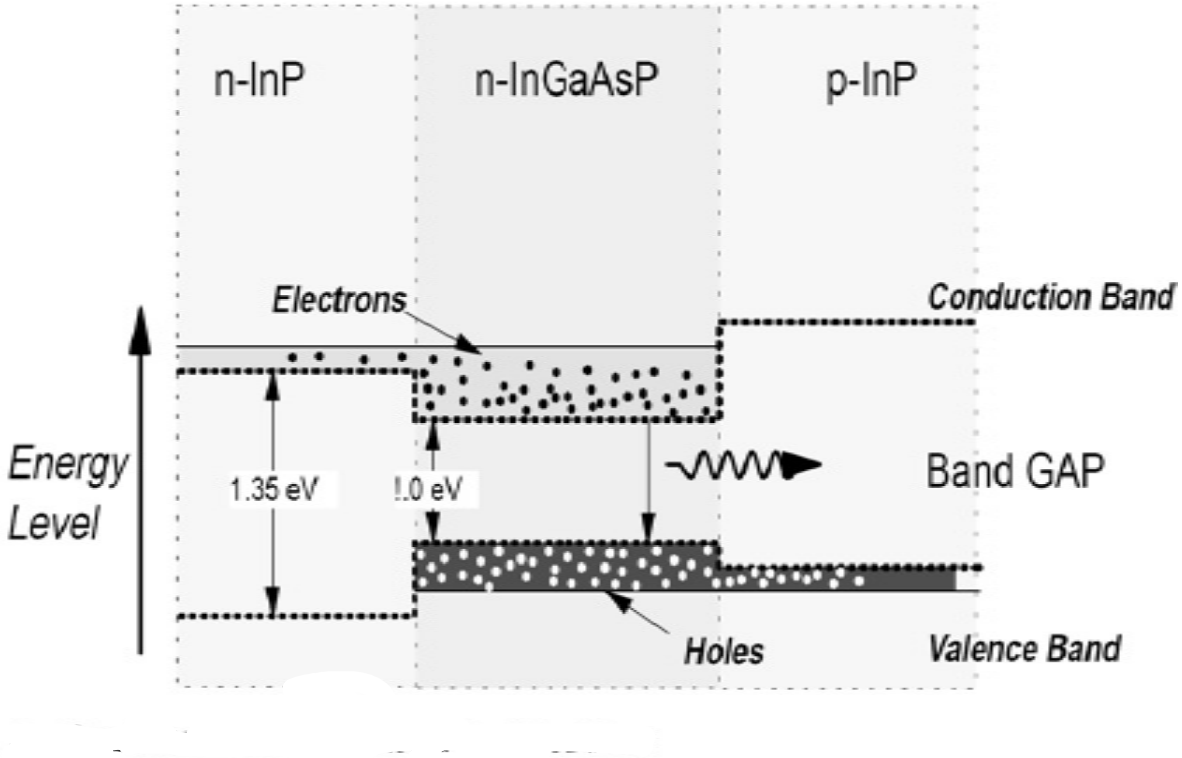






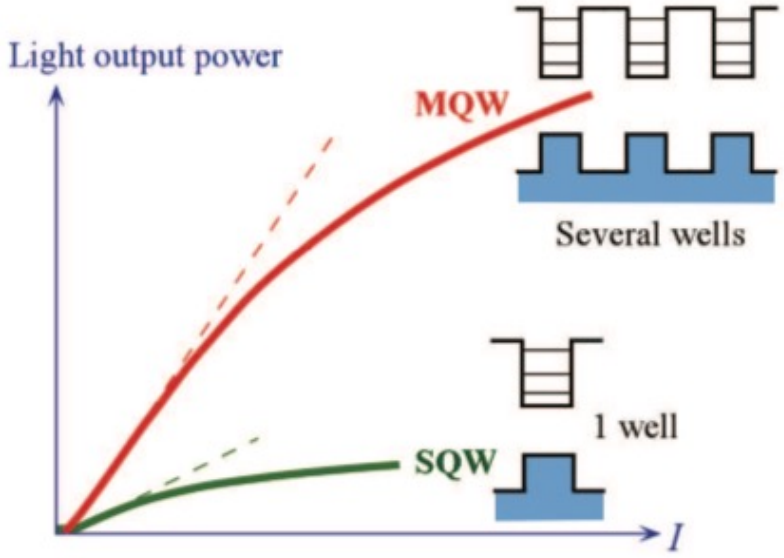
Improved LED Designs: Heterojunctions

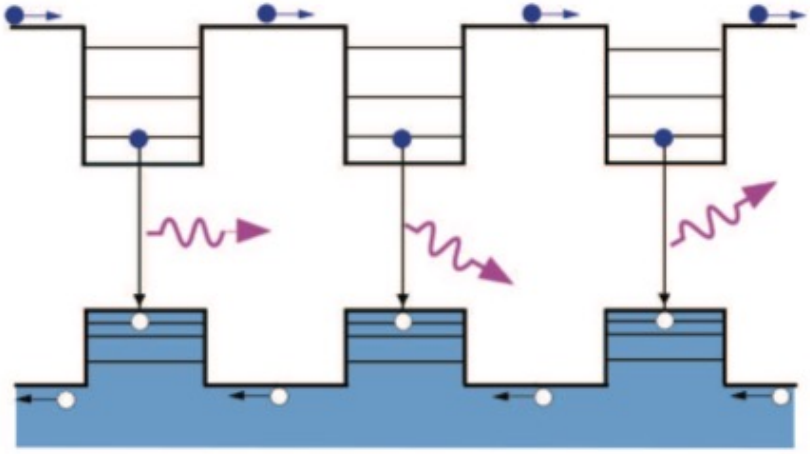
Improves carrier and optical confinement.



Improved LED Designs: Quantum Wells

Improves carrier and optical confinement.





Improved LED Designs: Quantum Wells

Example.

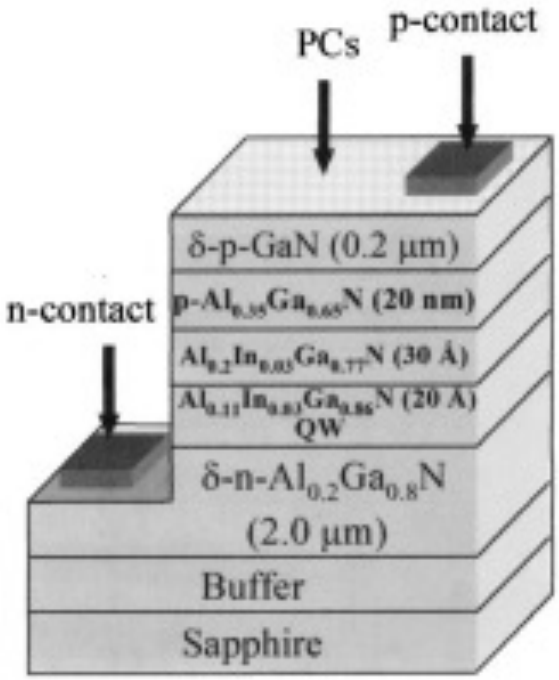


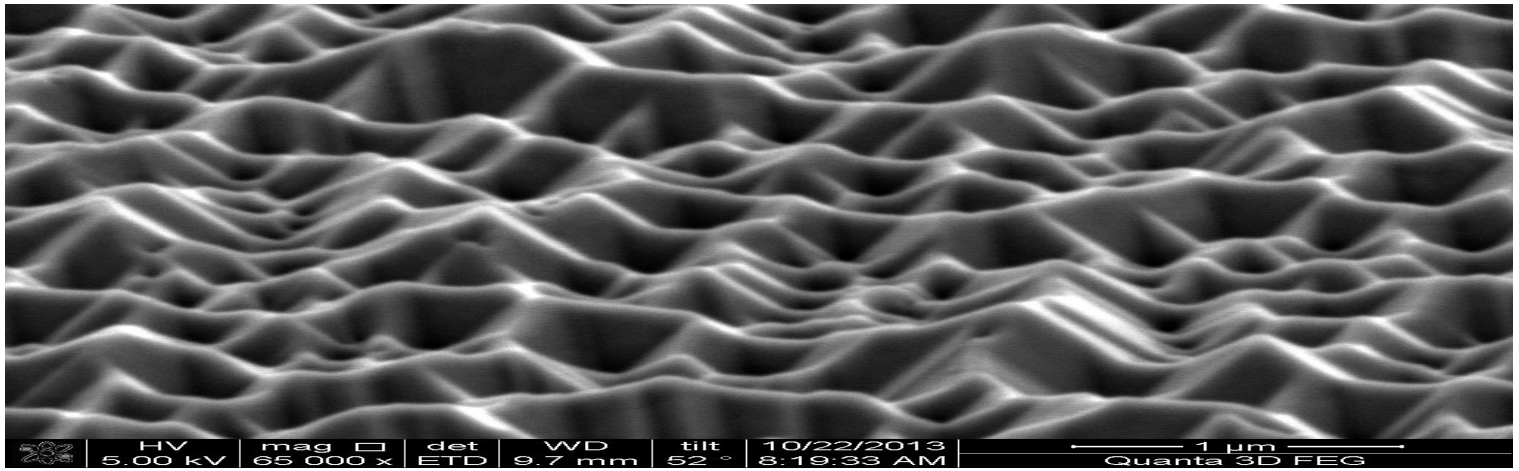
Table 4: The blue light emitting diode structure

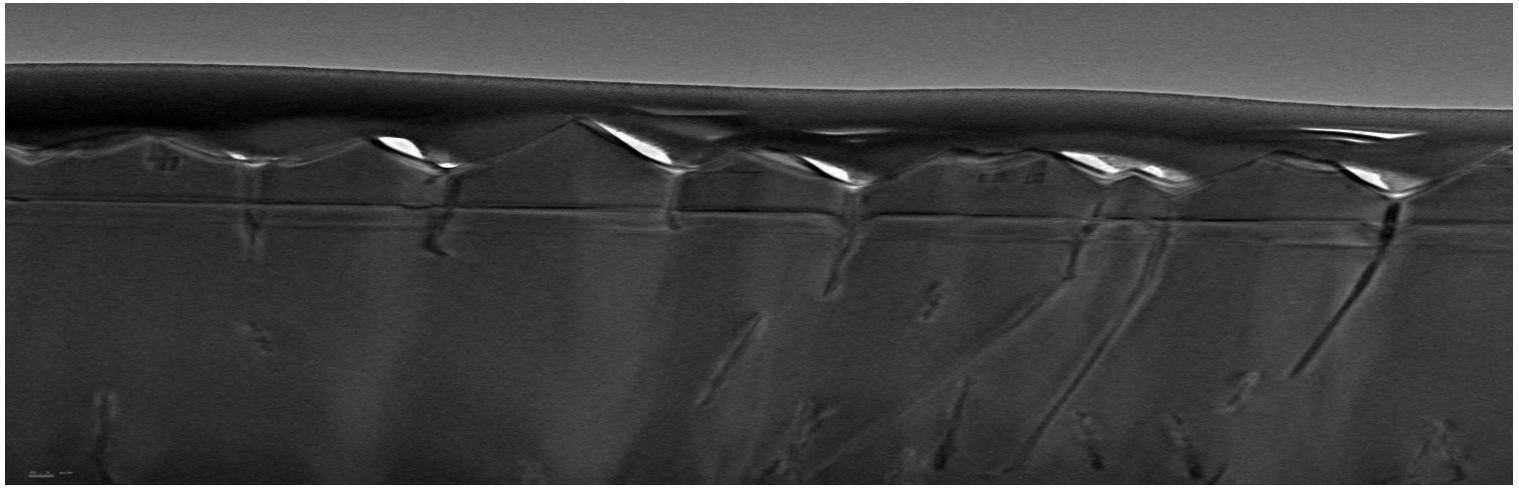
Material	Thickness	Dopant
GaN	1nm	Si, ~1^19
GaN	150nm	Mg, >1^19
Mg-EBL	100nm	Mg, ~5^18
InGaN/GaN (MQW)	300nm	UD
GaN	50nm	Si, >1^18
AlGaN	50nm	Si, >1^18
GaN	3um	Si, >1^18
GaN	1-2um	UD
Buffer		
Sapphire		

Table 5: The green light emitting diode structure.

Material	Thickness	Dopant
InGaN	2nm	Si, ~1^19
GaN	600nm	Mg, >5^18
Mg-EBL	50nm	Mg, ~5^18
InGaN/GaN (MQW)	200nm	UD
GaN	3um	Si, >1^18
GaN	1-2um	UD
Buffer		
Sapphire		

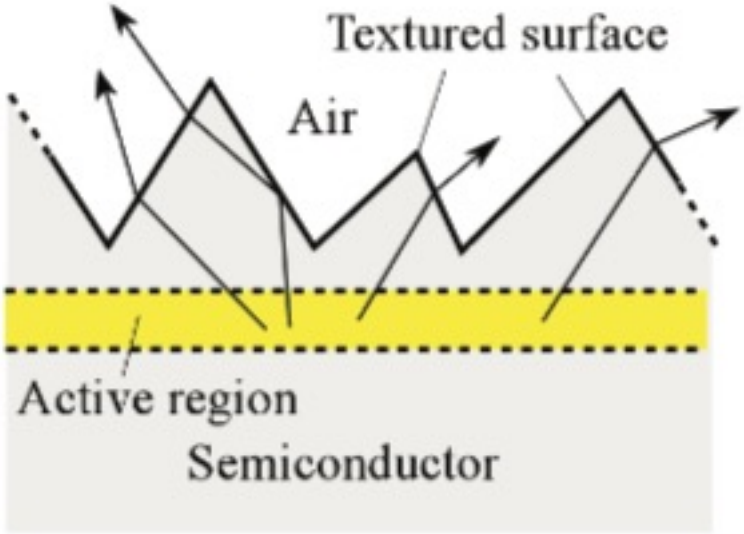
Improved LED Designs: Quantum Wells

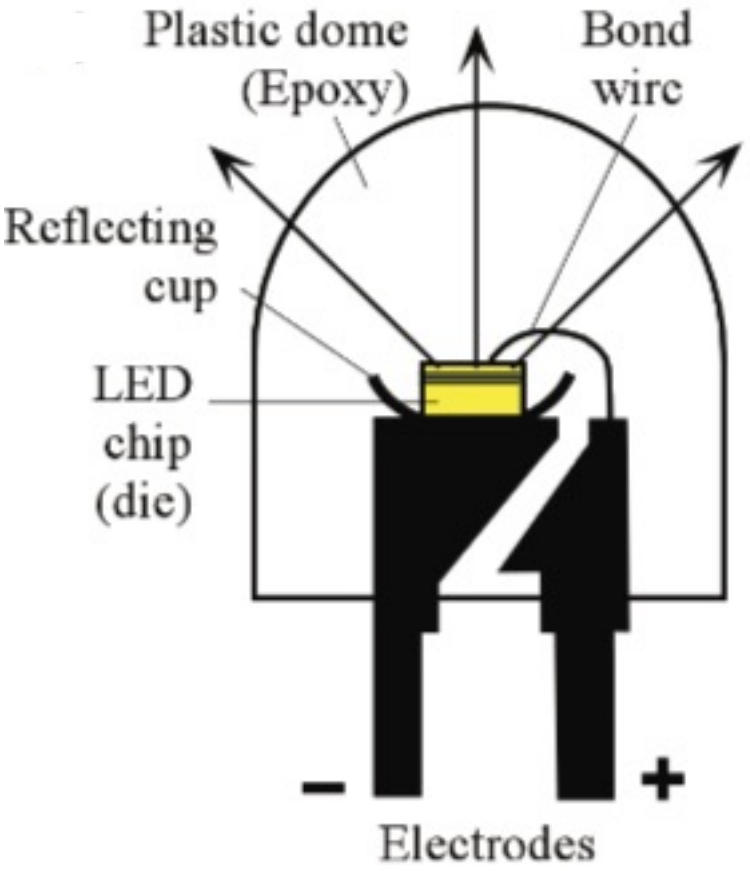




Improved LED Designs: Encapsulation

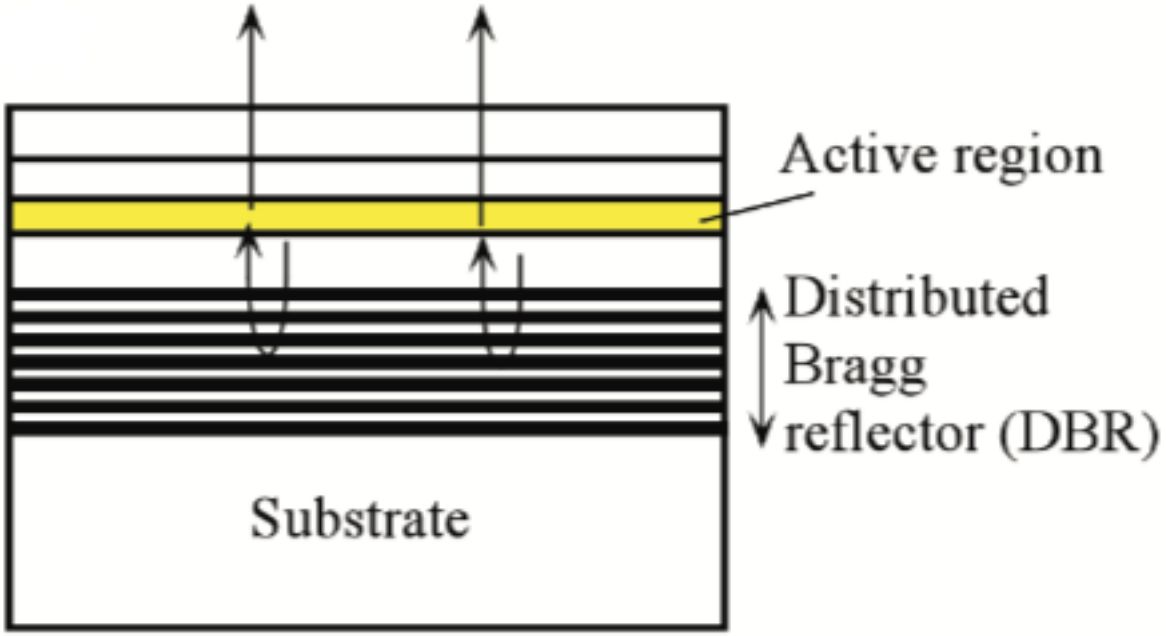
Improves light Extraction from LED.



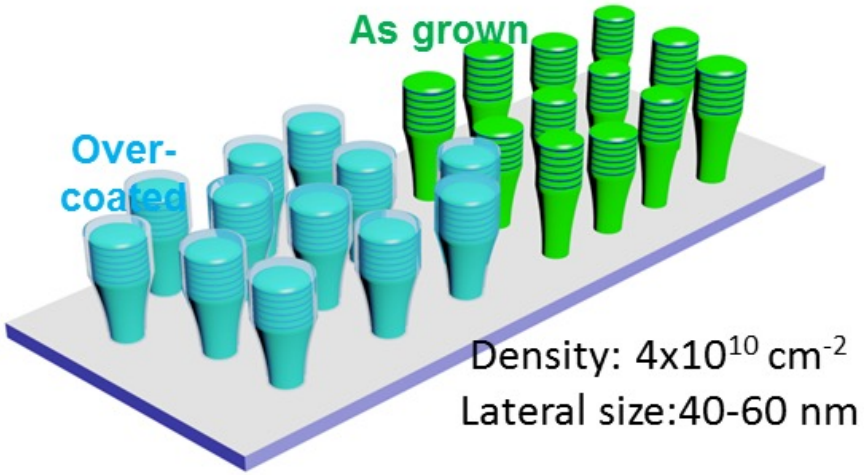


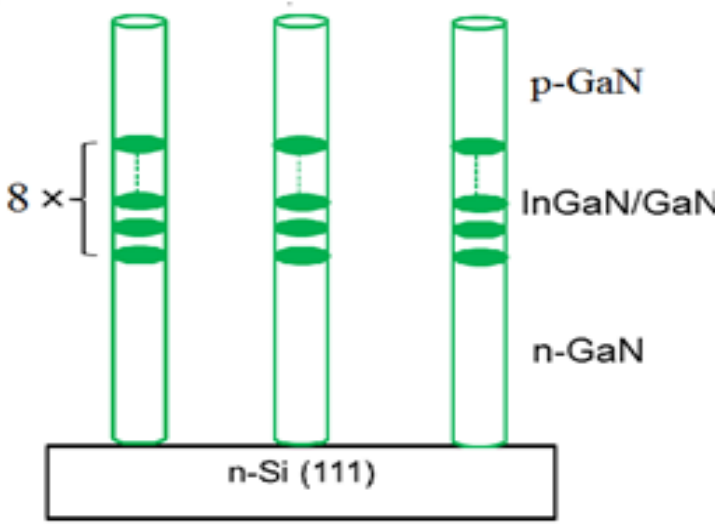
Improved LED Designs: Mirror

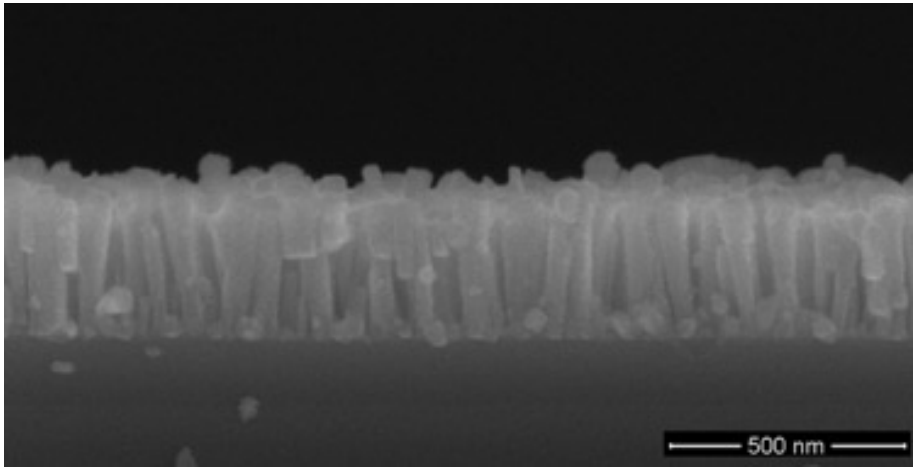
Improves light Extraction from LED.

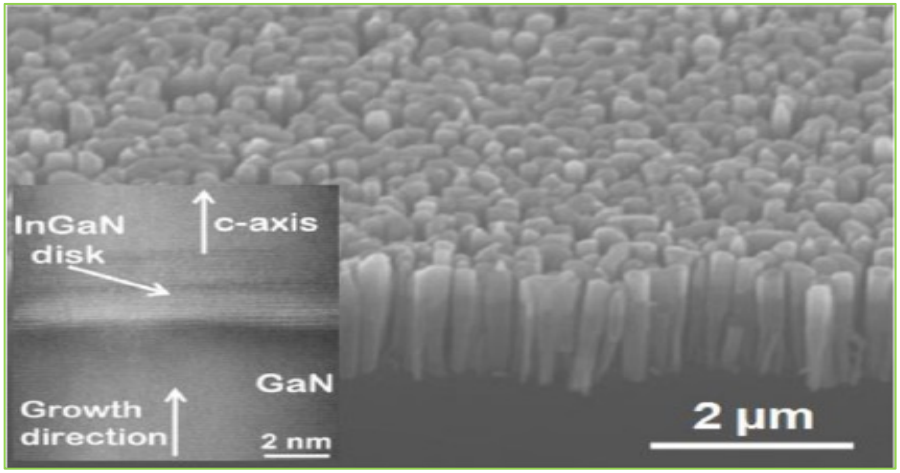


Improved LED Designs: Nanophotonics









Improved LED Designs: Nanoplasmonics

Nano Photonics and Plasmonics Research

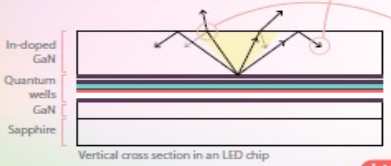
Effecient Light Extraction from LEDs

An Experimental Research-Based Senior Project
Electrical Eng. Dept. - King Fahd University of Petroleum & Minerals

Problem Definition

95% of light generated inside an LED is wasted!

This is because of 2 reasons:



Vertical cross section in an LED chip

1. **Total Internal Reflection (TIR)**
Passing through LED-air interface is limited to light arrays coming at an angle within the escape cone. The rest are reflected.

2. **Partial Reflection**
Only part of the light coming at an allowed angle passes. The rest is also reflected.

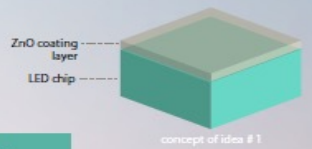
we investigated **3 IDEAS** to solve this problem

our objective is to increase the light extraction effeciency (LEE) of our GaN LED chips by manipulating them using first principles from electromagnetics. In other words, we want to minimize reflection and widen the escape cone.

IDEA # 1

Surface Coating

Use single-layer coating to reduce reflected light

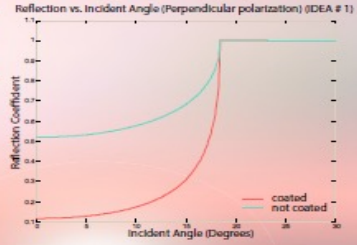


concept of idea # 1

from literature:
- best layer thickness $= \frac{\lambda}{4}$ (wavelength of color emitted from LED in coating material)
- best coating material should have $\epsilon_{r\,coat} = \sqrt{\epsilon_{r\,air}\,\epsilon_{r\,LED\,top\,layer}} = 3.16$ in our case (relative permittivity)

The closest material we could get was zinc oxide (ZnO) of $\epsilon_r = 4$

Simulation Results for ZnO coating
- Significant decrease in reflection :)
- No change in escape cone :(

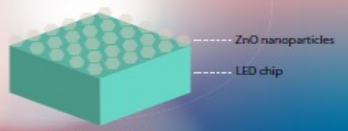


IDEA # 2

Surface Nanoparticles

Randomly scatter ZnO nanoarticles on surface of LED
This can reduce TIR by making more light encounter the LED-air interface at a normal (or semi-normal) angle

Experimental Results
- No significant difference in light emission :(
- Since this is not expected, this experiment should be repeated

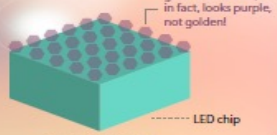


concept of idea # 2

IDEA # 3

Gold Nanoparticles

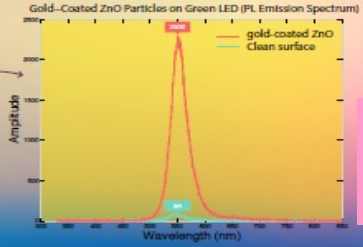
Randomly scatter gold-coated ZnO nanoarticles on surface of LED
This has the benifit of idea # 2 + the plasmonic effect from electrons in gold



concept of idea # 3

Experimental Results

24 times more light emission! =)
- This is a large increase in light emission :)
- From our literature review, we think no budy have deposited gold-coated ZnO on an Indium-doped GaN-based LED before :)



Project Conclusions
- Single-layer coating, when properly done, can significantly increase LEE
- Au-coated ZnO nanoparticles can increase light emission up to more than 24 times

Team Advisor : Dr. Mohammad Al-Sunaidi

Team Members

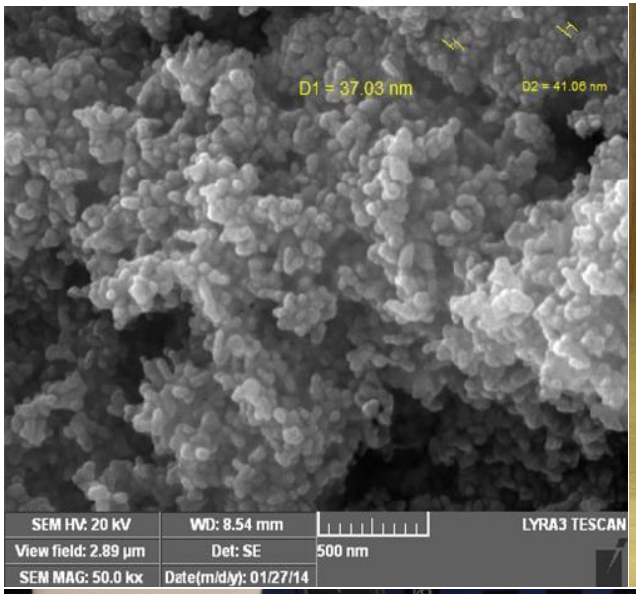
Our sincere thanks to:
Dr. Mohammad Gondal (Physics Dept.)
Dr. Mazen Khaled (Chemistry Dept.)
Dr. Zain Yamani (CENT)
Dr. Boon Ooi (KAUST Photonics Lab)
Mr. Rashid Siddique (CENT)
Dr. Mohd Sharizal (KAUST Photonics Lab)
Dr. Muhammad Haider (Physics Dept.)
Mr. Emad Alkhazraji (Grad. Student, EE Dept.)

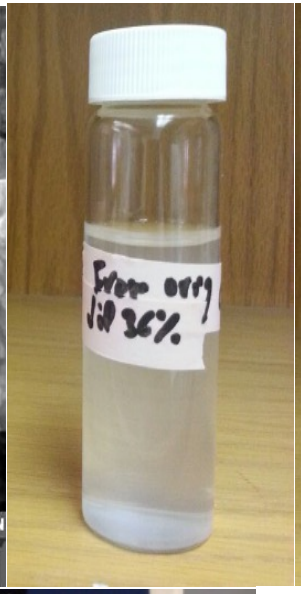
Designed by @ThinkerAli May 2015 Background credit for freepik.com

2015 IEEE Conference on Applied Electrical Engineering and Computing Technologies (AEECT Nov, 2015)

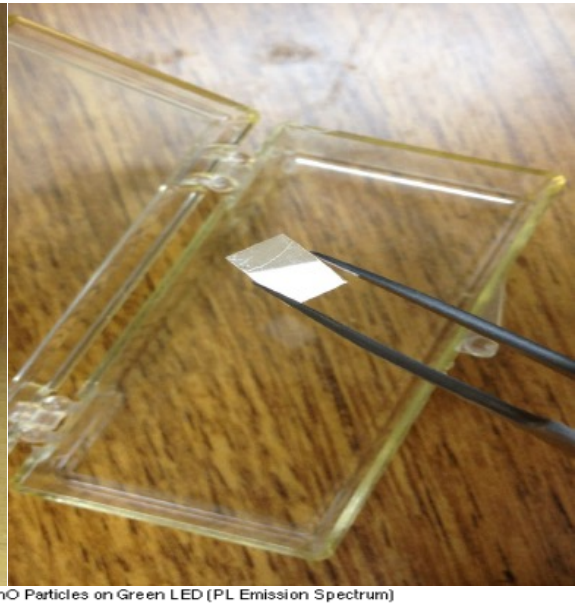
Improved LED Designs: Nanoplasmonics

Nano Photonics and Plasmonics Research

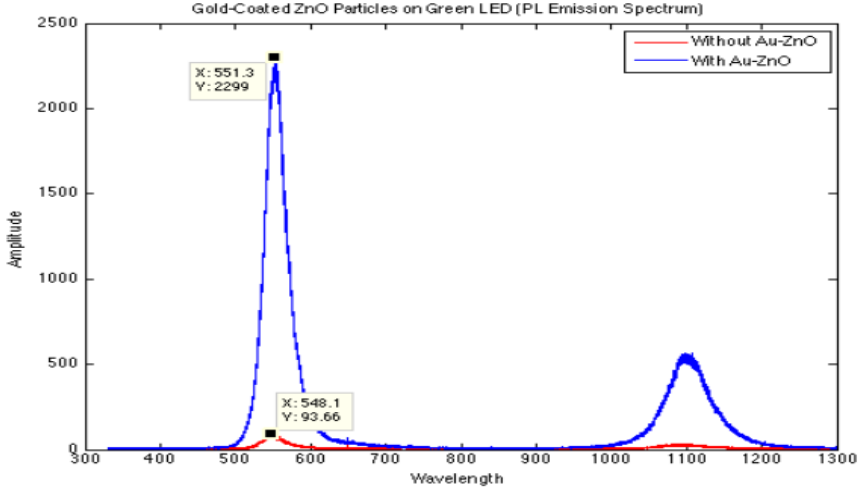




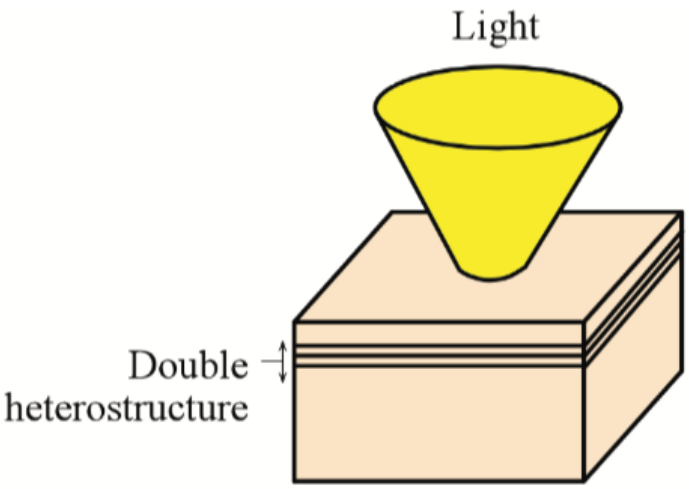




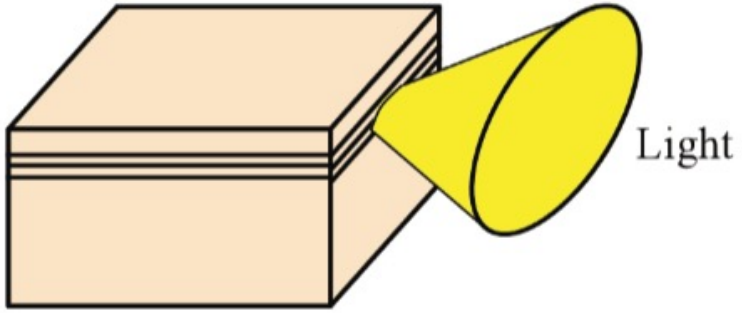




Basic LED Types



Surface emitting LED



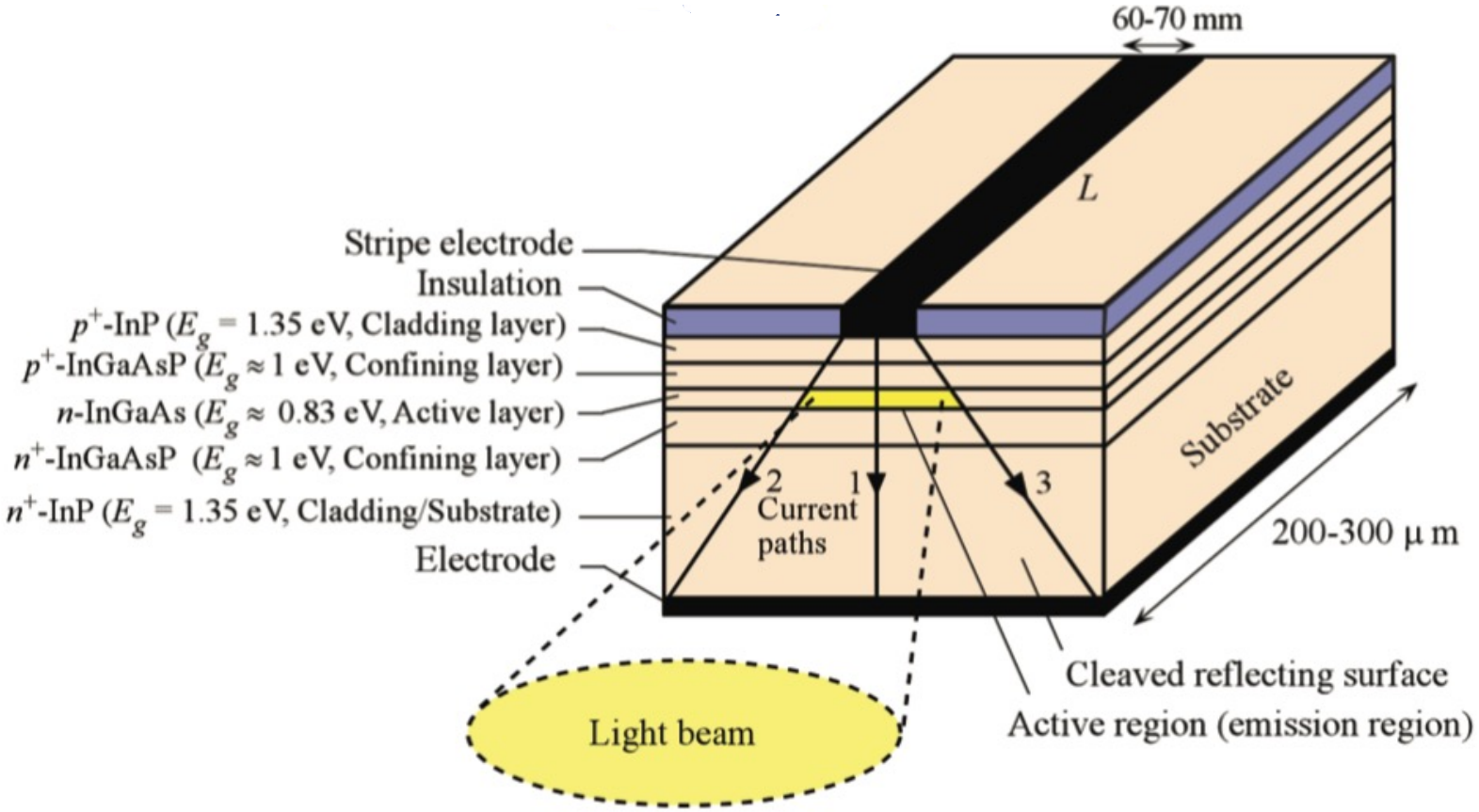
Edge emitting LED

Lighting
Display
Applications

Optical
Communications

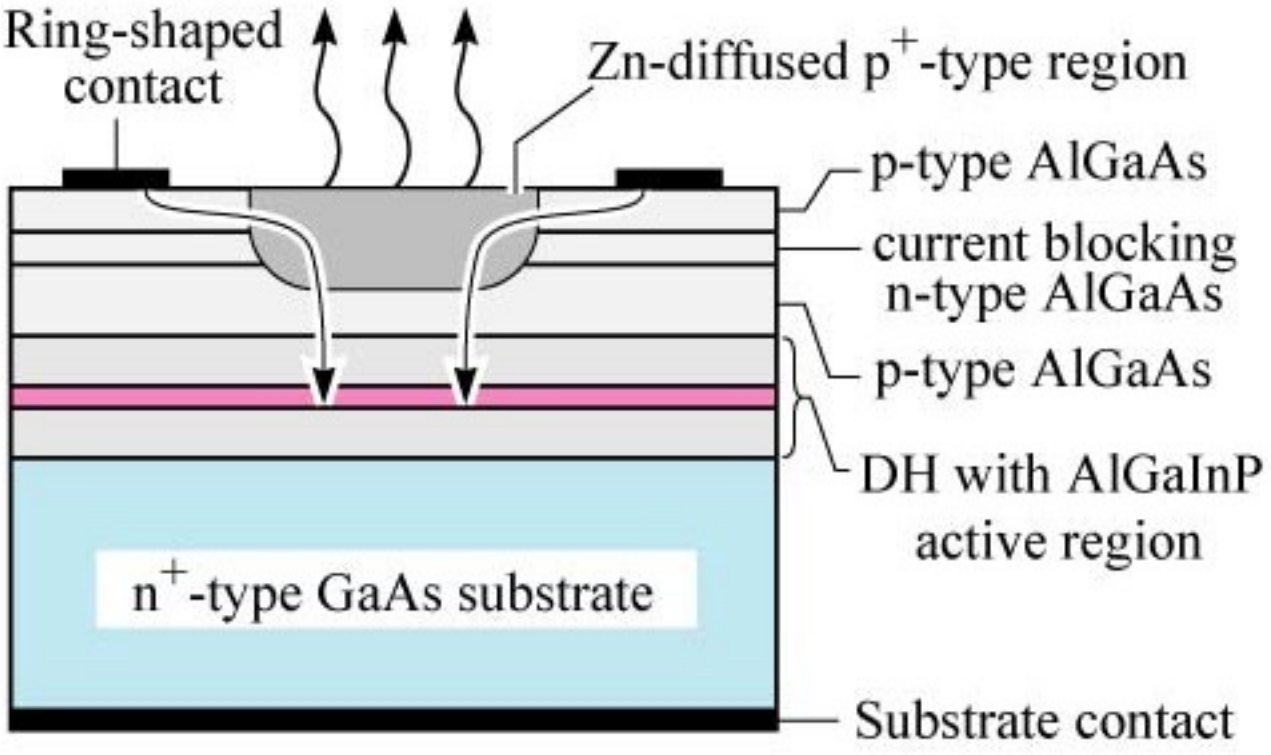
Edge-emitting LED

Example.



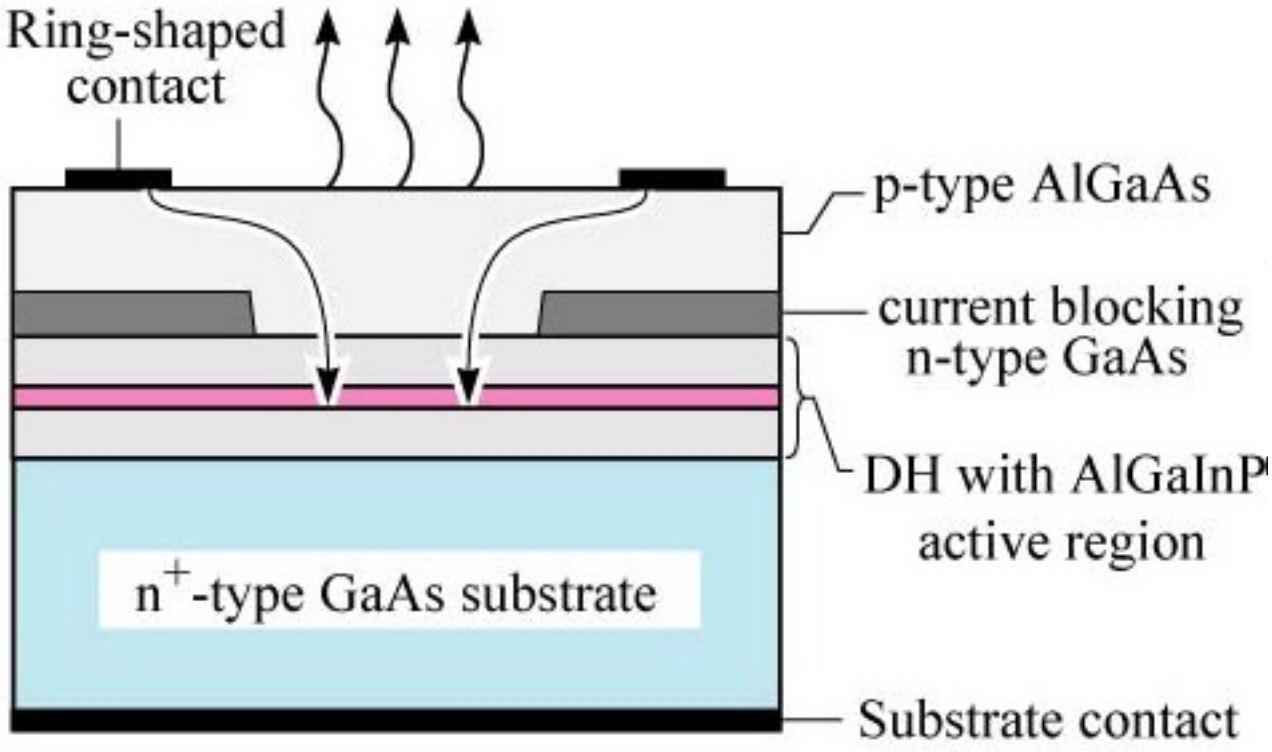
Surface-emitting LED

Example.



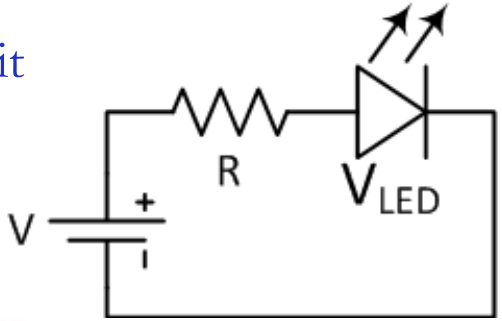
Surface-emitting LED

Example.

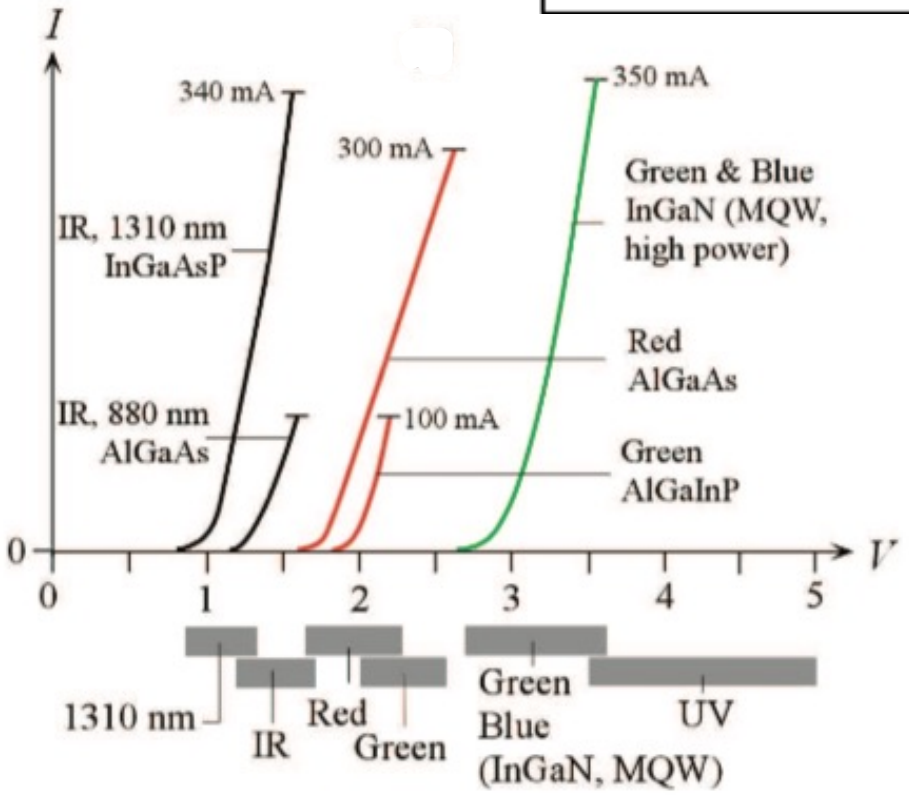


LED Characterization: Output Optical Power

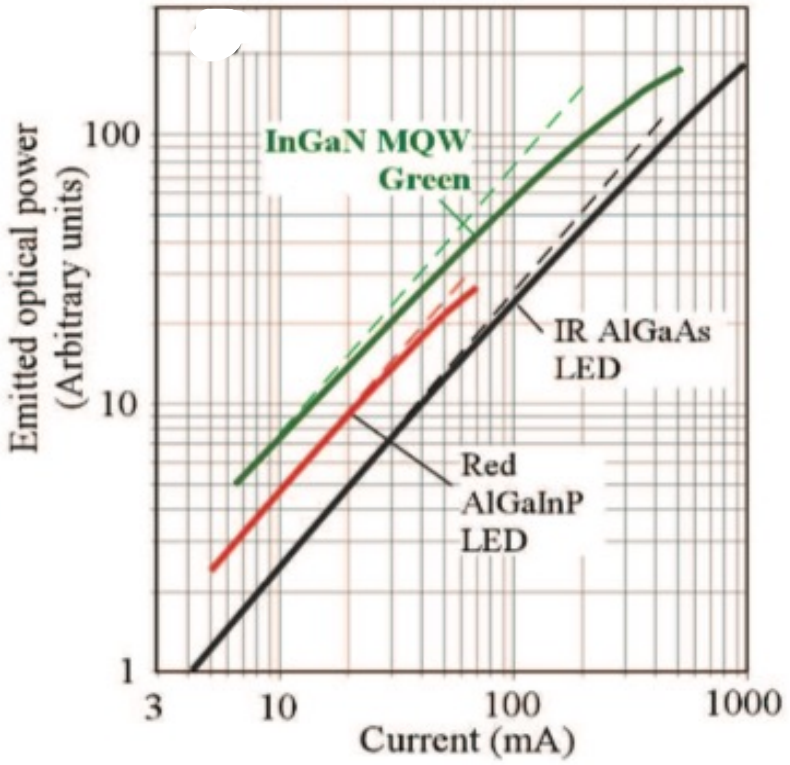
Circuit



IV Curve



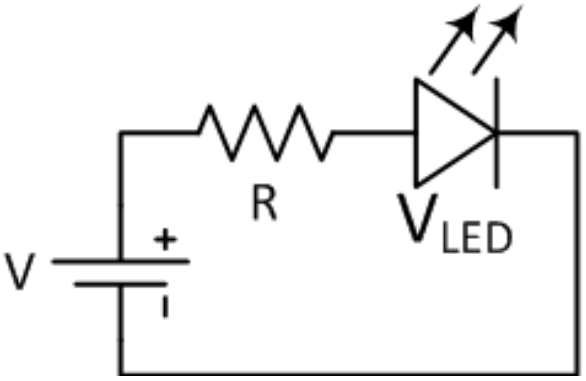
PI Curve $P_o = \hbar\omega\, \eta_{ex}\, \frac{i}{q}$

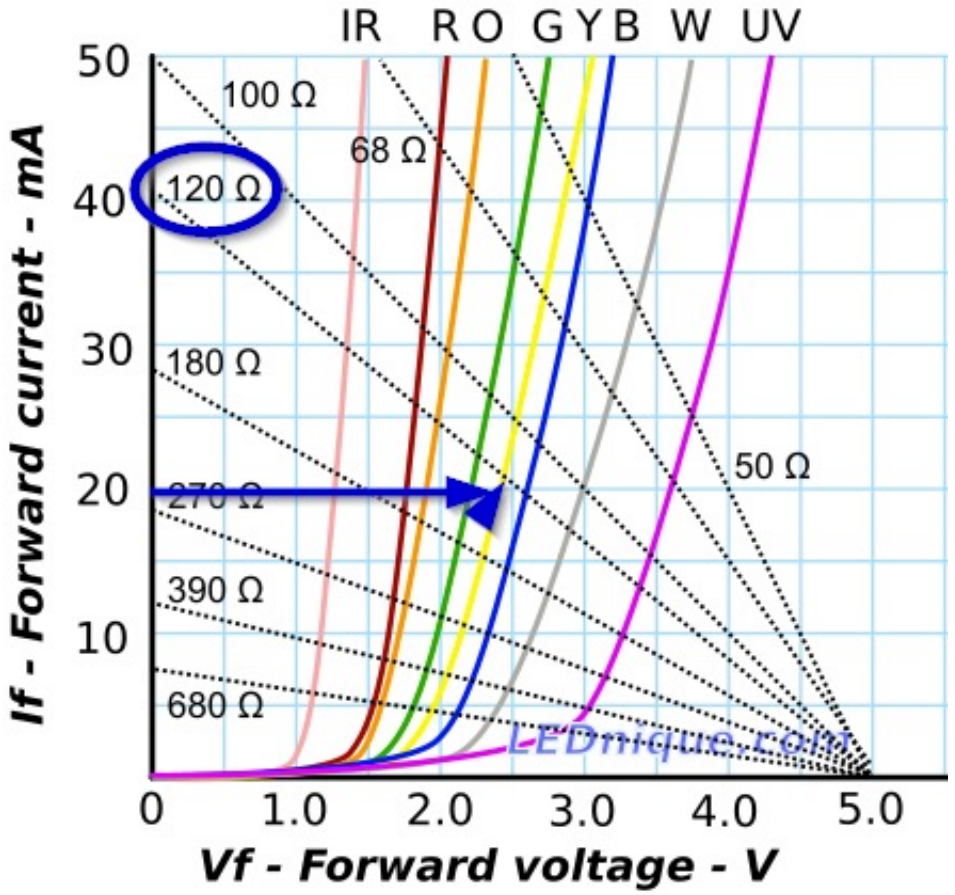


- Carrier leakage
- Joule heating

LED Characterization: Output Optical Power

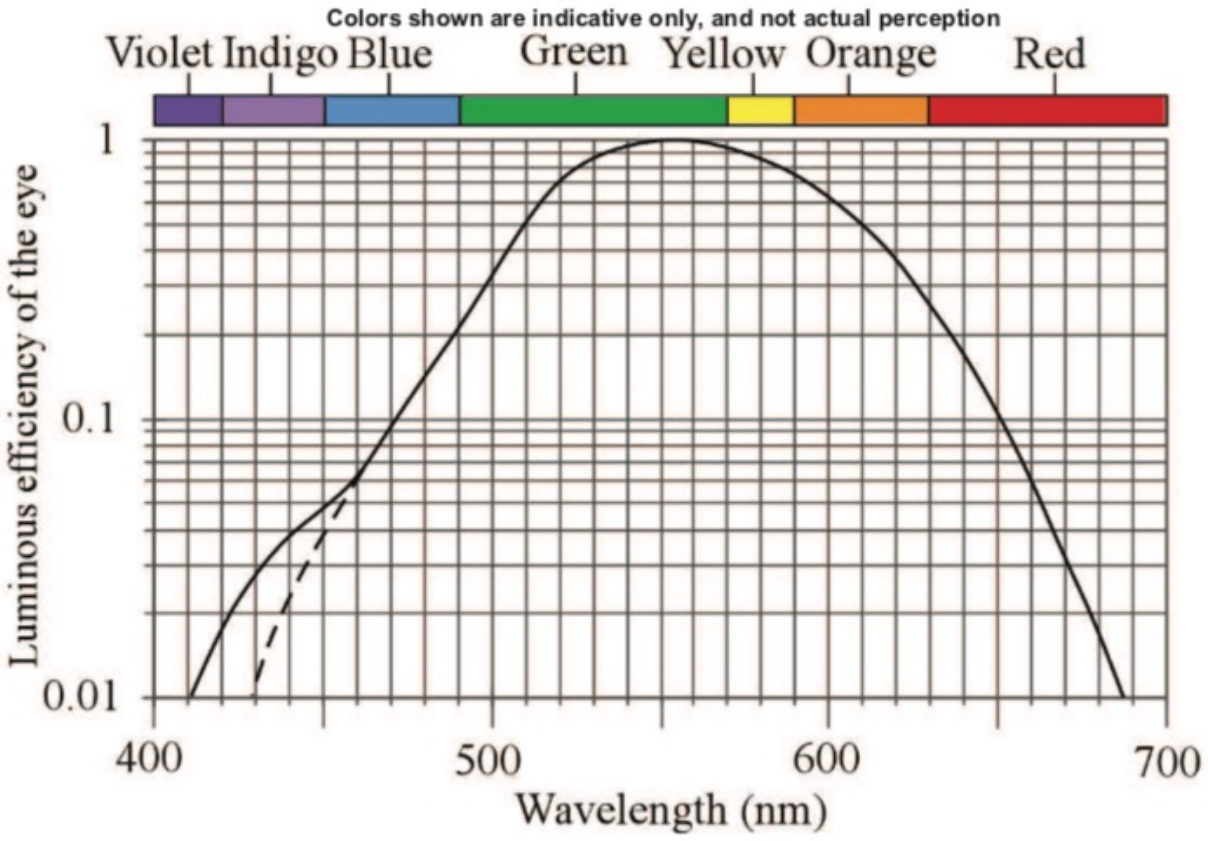
Example: Find the value of the series resistance for a yellow LED biased at 20 mA. Take V=5 V.





LED Characterization: Output Optical Power

The human eye optical spectrum: Luminous efficiency.



Example: What is the luminous flux of a red LED emitting at 650 nm with output power of 10 mW?

Luminous efficiency at 650 nm is 0.1.

Luminous flux is given by

$$Lum = \eta_{LE} P_o (633\ lm/W)$$

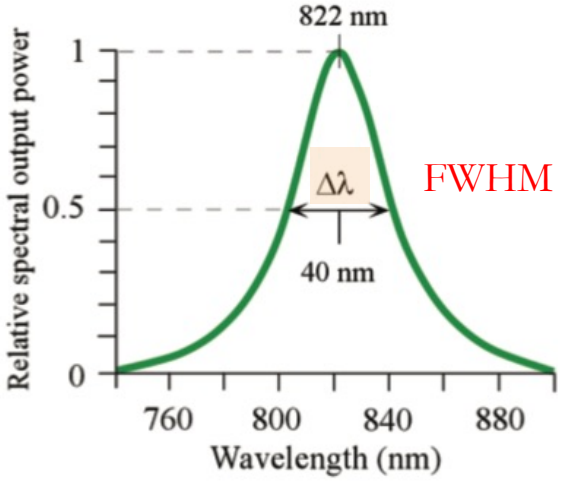
$$Lum = 0.63\ \text{lm}$$

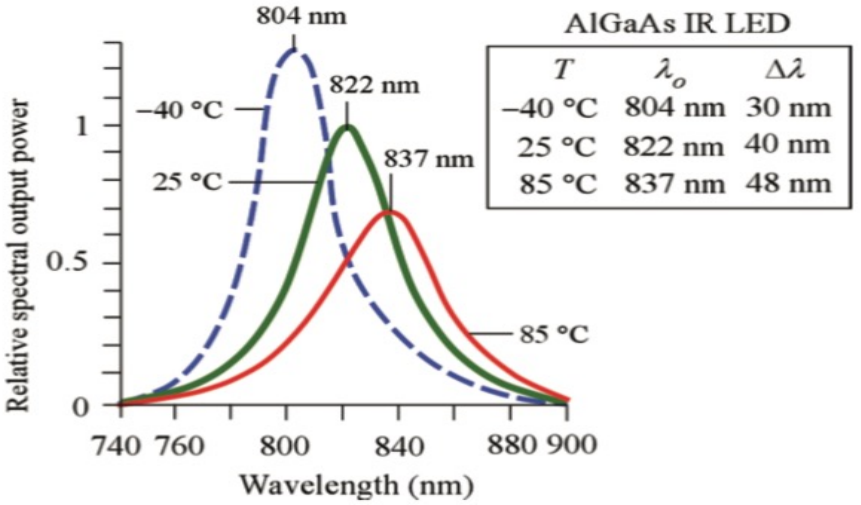
Repeat for green at 532 nm.

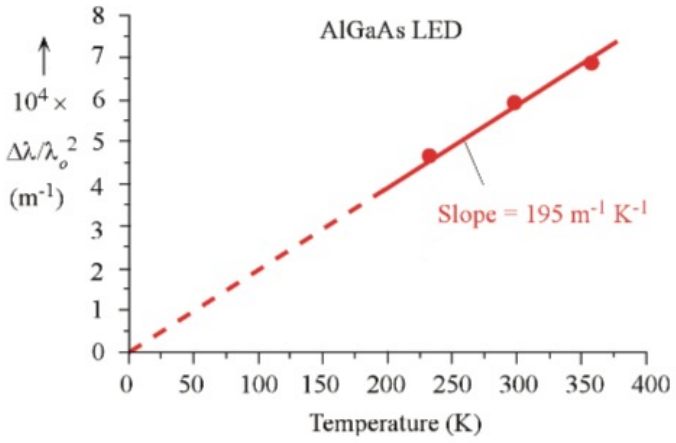
➡ $Lum = 5.51\ \text{lm}$

LED Characterization: Spectral Distribution

Spectral Distribution of an LED is described by peak wavelength λ_o and spectral width $\Delta\lambda$.







$$E_G = E_{Go} - \frac{AT^2}{(B+T)}$$ Varshni Equation

- Red-shift with temperature is due to decrease in bandgap.
- Increase of spectral width with temperature is due to electron-hole distribution in the conduction/valence bands.

LED Characterization: Spectral Distribution

We can express the spectral width $\Delta\lambda$ in term of change in photon energy $\Delta\hbar\omega$.

$$\lambda = \frac{hc}{E}$$

$$\frac{\Delta\lambda}{\Delta E} = hc\left(\frac{-1}{E^2}\right)$$

$$\Delta\lambda = -hc\left(\frac{\Delta E}{E^2}\right)$$

$$\Delta\lambda = -\left(\frac{\lambda^2}{hc}\right)\Delta E$$

$$\frac{\Delta\lambda}{\lambda^2} = -\frac{\Delta E}{hc}$$

$$\Delta E \approx 2.5 - 3\,k_B T$$

Example: Find the spectral width for a GaAsP LED emitting at 655 nm at 300 K. Take $\Delta E = 3\,k_B T$.

$$\Delta\lambda = \left(\frac{(655\times10^{-9})^2}{6.626\times10^{-34}\times3\times10^{8}}\right)(3\times1.38\times10^{-23}\times300)$$

$$\Delta\lambda = 26.8\,nm$$

Example: Find the shift in the emitted wavelength of a GaAs LED per 1 °C change at room temperature. E_{Go}=1.519 eV, A= 5.41×10^{-4} eV/K, B=204 K.

$$\frac{\Delta E_G}{\Delta T} = -\left(\frac{E_{Go} - E_G}{T}\right)\left(\frac{T + 2B}{T + B}\right)$$

Q & A

Send me an email if you have any questions.